What I Wish I Knew in First Year

Ruán Murgatroyd

Who am I?

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Former CS++ Chair & Sysadmin

First Class Honours Graduate

Competitor @ WorldSkills Shanghai 2026

Former Reserve for Irish Cybersecurity Team

TU858 Alumnus

SAP Grad & former SAP STAR Student

Father of Juno and Legs

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JUNO (Princess)

Who am I?

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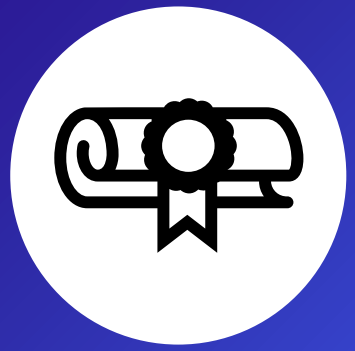
ybersecurity Team

AR Student

LEGS (Evil)

JUNO (Princess)

Content



UNIVERSITY

- Tips
- Need to know

CAREER

- What employers look for
- Internships
- Where to look

Content

LIFE

- Why are you here?
- Setting goals
- General advice

AI

- AI in university
- AI in work
- How I use AI

UNIVERSITY

Being Proactive

SELF CARE

- You’re here for you
- Only you know how you’re doing
- Look out for yourself

GETTING HELP

- Know how to identify when you need help
- Ask for it EARLY
- Friends, lecturers, Academic Hub

LEARNING

- Learn what you need
- Only you know how you’re doing
- Look out for yourself
- **ASK EARLY**

You don’t need all 100%

Software development tips

Want to code better?
CODE MORE

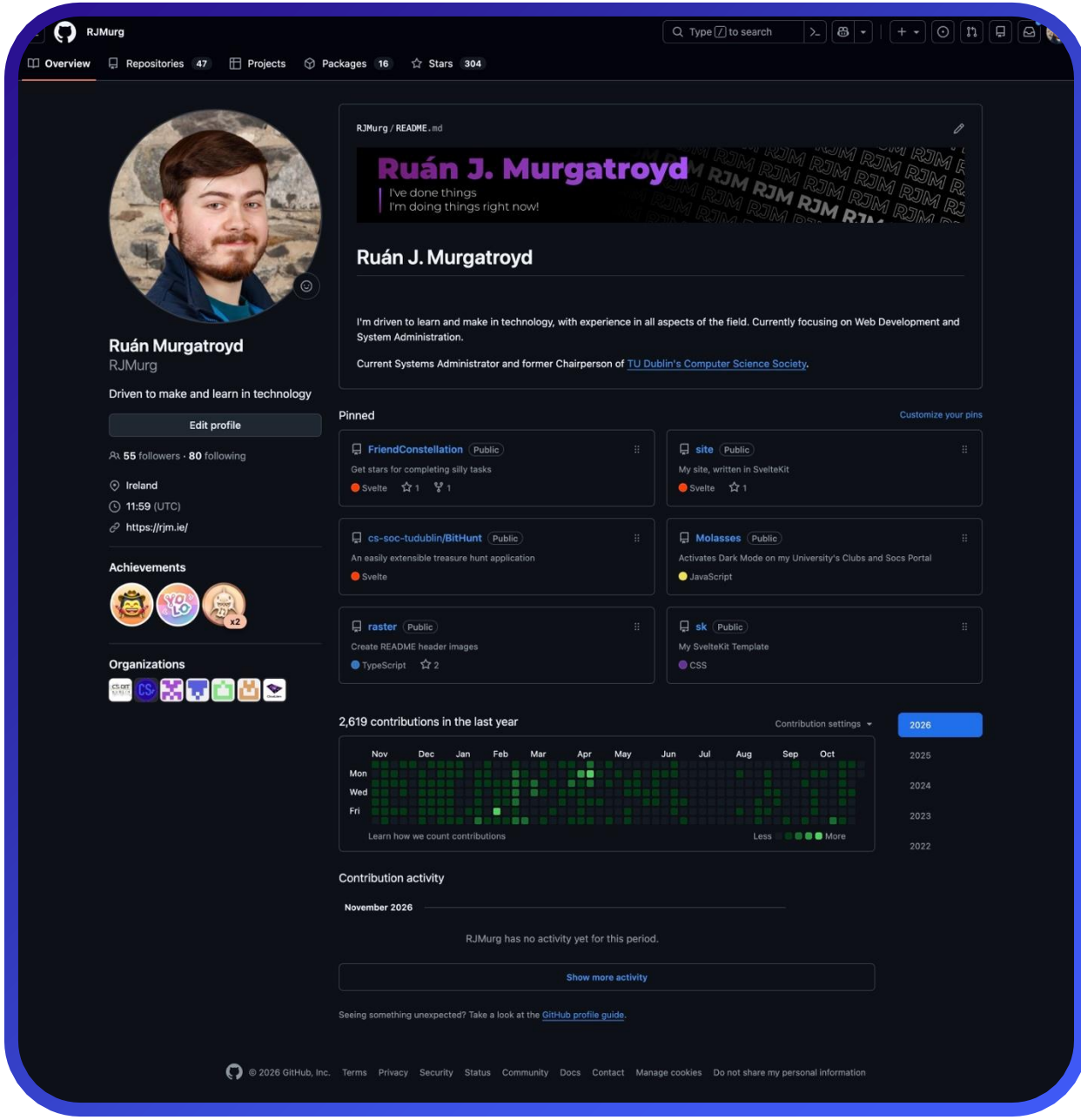
Learn the fundamentals

Learn from others &
Try new things

Turn your ideas into code

Git & GitHub

- Git is the **ONLY** standard in the industry
- Good for personal & college projects
- GitHub is like LinkedIn for devs
- Employers **WILL** look at it, if you link it



CAREER

What do employers look for?

TECHNICAL SKILLS

- Your ability to program
- Your ability to understand
- Your ability to architect
- Your ability to test & secure

SOFT SKILLS

- Leadership
- Communication
- Problem-solving
- Time management

THE BIG THREE

ENTHUSIASM

- Interested in what you do
- Striving to get better
- Wanting to help others

How to prove them

- Relevant work experience
  - Non-tech can still be useful
- Education
  - A prerequisite, but everyone will have this. Some might have better (Masters, PhD)
- Self-Learning
  - Personal projects, open-source contributions, quantifiable work outside the classroom
- Volunteering
  - Clubs, Societies, Charities & Entrepreneurship

Personal projects should

Be simple

Don’t overcomplicate it, don’t make it solve every problem

Solve a problem

Even better if it solves problems for others too

Let you learn

Employers will ask you, waste of time to do something & learn nothing

Be something you know

Employers will ask, describe it!

LinkedIn

- **GET IT**
  - Include a photo
  - Keep it mostly up-to-date
- What it shows:
  - You're not lazy
  - Your posts show your personality, interests & can prove the big 3
- What to do:
  - Don't spend too much time on it
  - Connect with *everyone*

Pre-3rd year internships

- SAP STAR
- Google STEP
- & More

~~**“What if I’m not good enough?”**~~

LIFE

Why are you studying?

- Just for the degree?
- To get job ready?
- To learn?
- To connect?
- To enjoy yourself?

Goal setting

- Pick a goal
- Make a plan
- Act

Other advice

- Don’t feel like you’re behind the ‘class geniuses’
- Don’t be afraid to fail, you will learn
- Consistency > everything
  - Make it fun, or you won’t enjoy it
- Mind your health. Mental & physical
- Step out your comfort zone
  - The worst they can say is no
- Enjoy your time in Uni
  - You’ll never be this young again

AI

AI in University

- TU Dublin *knows* you will use AI
- TU Dublin **forbids** the use of AI for completing your assignments / exams (Unless stated otherwise)
- You are paying ~2,500 (OR MORE) just to get Claude to do your work?
- You will learn **NOTHING** if you don't engage

AI in Employment

- Every employer knows you can use AI
- Employers want people who *understand* code, architecture, security and optimisation
- If your entire portfolio is vibe-coded, and you can’t talk about why you did what you did, employers won’t be interested

How I used AI in University

- An assistant to help me understand
- Offer critique or further analysis on a topic
- Minor refactors
- Architecture reviews

Q&A

No question is too small.

Questions from Previous Years

- What are the best projects for learning to program?
- How do you organise your coding environment?
- How did you manage comparing yourself with others?
- How do you manage your time?

Eisenhower Matrix

Thank you!

Join us on Discord:
discord.cspp.ie

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